LESSON 2.2

Represent and Describe Data in a Histogram

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make and read a histogram to display data in intervals.

WHY THIS MATTERS The basketball league has 30 players and wants to display everyone's points-per-game average in a way that shows how the data is distributed. The raw averages are: 2, 5, 7, 8, 9, 10, 11, 11, 12, 13, 13, 14, 14, 15, 15, 16, 16, 17, 18, 18, 19, 20, 21, 22, 23, 24, 25, 28, 30, 32. Listing all 30 numbers is hard to read — a histogram will reveal the pattern!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Histogram *Spanish: Histograma*	A bar graph that groups data into equal ranges. The bars touch.	Bars that touch
Frequency *Spanish: Frecuencia*	How many times a value shows up.	Bar height shows the count
Interval *Spanish: Intervalo*	A range of numbers used to group data.	10-19 is one interval
Distribution *Spanish: Distribución*	How the data is spread out.	Tallest bar in the middle
Variability *Spanish: Variabilidad*	How spread out the numbers are.	Data in just 2 intervals = low variability. Data across 6 intervals = high variability
Frequency table *Spanish: Tabla de frecuencias*	A table that lists each interval and how many values fall in it.	10-19 . . . 3

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Walk times (min): 4, 15, 9, 22, 11, 18, 27. Use 0–9, 10–19, 20–29.

1. 0–9 holds 4 and 9: frequency 2.
2. 10–19 holds 15, 11, 18: frequency 3. 20–29 holds 22, 27: frequency 2.
3. Draw touching bars of height 2, 3, 2.

Answer: Check: 2 + 3 + 2 = 7 values

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

New data: 2, 4, 8, 11, 13, 16, 21. We will use intervals of 10 again: 0–9, 10–19, 20–29.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- My intervals are _____ wide and they do not _____ .
- The interval _____ holds _____ values.
- The tallest bar is _____ because _____ .
- The bars touch because the intervals _____ .

Word bank histogram · interval · frequency · tally · equal · overlap · bar · height · count · touch · distribution · data

WATCH OUT!

A common mistake

A common mistake in Display Data: Histograms is counting a boundary value in two intervals at once

LESSON 2.2

Represent and Describe Data in a Histogram

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

In a histogram, what does the height of each bar represent?

(A) The frequency (count) of data in that interval

(B) The average of the data in that interval

(C) The range of the interval

(D) The median of the data

HINT Think about what you'd count to draw one bar: you'd tally how many data values land inside that interval.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Order the steps for making a histogram.

Groups: Steps (in order)

_____ Step 1: Choose equal-sized intervals that cover all the data

_____ Step 2: Count how many data values fall in each interval (frequency)

_____ Step 3: Draw bars for each interval — height = frequency

_____ Step 4: Make sure bars touch with no gaps

HINT You must decide on intervals before counting data in each one.

3 **Circle the letter of the best answer. Show how you know.**

A histogram bar for the interval 20–29 reaches a height of 7. What does the 7 tell you?

(A) The largest value is 7

(B) There are 7 intervals

(C) The interval is 7 wide

(D) 7 data values fall between 20 and 29

HINT Two numbers appear here: the interval (20–29) and the height (7). Keep straight which one is a range and which is a count.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A histogram shows these frequencies: 0–4: 3 players, 5–9: 8 players, 10–14: 12 players, 15–19: 5 players. How many players are represented in total?

(A) 12

(B) 15

(C) 20

(D) 28

HINT The question asks for the TOTAL across all four intervals — not the tallest bar and not the biggest interval.

PER 1

1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

5 **Circle the letter of the best answer. Show how you know.**

How is a histogram different from a regular bar graph?

(A) Histogram bars touch (no gaps) because the intervals are continuous ranges

(B) There is no difference

(C) Histograms use colors and bar graphs don't

(D) Bar graphs show numbers and histograms show words

HINT Picture both graphs side by side: one has spaces between its bars, one doesn't. Ask why that would be.

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A histogram has intervals: 0–4, 5–9, 10–14, 15–19. What is the width of each interval?

(A) 4

(B) 5

(C) 10

(D) 19

HINT Width means how many whole-number values fit inside one interval — count them, don't just subtract the endpoints.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The tallest bar is the 10–19 interval with 16 players. What does the shape of this histogram tell you about scoring in the league?

SENTENCE STARTER Most players score between ___ and ___ points per game. The histogram is skewed ___ because ___. This tells me that ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.2

Represent and Describe Data in a Histogram

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Interval (Yards)	Frequency
Player rushing yards per game: 0, 5, 12, 15, 18, 22, 25, 28, 31, 35, 40, 45	
0–9	
10–19	
20–29	
30–39	
40–49	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each term to its description.

_____ Histogram
_____ Bar graph
_____ Frequency
_____ Interval

(A) Bars touch, shows frequency by interval
(B) Bars separated, shows categories
(C) Count of values in an interval
(D) Range of values grouped together

3 **Write the name of the correct group on each line.**

Rank the intervals from highest to lowest frequency.

Groups: Most to fewest players

_____ 10–19: 11 players

_____ 20–29: 7 players

_____ 0–9: 4 players

_____ 30–39: 2 players

4 **Complete the table. Show how you found each value.**

Use this data set of rebounds per game to complete the frequency table (intervals of 5): 1, 3, 4, 6, 7, 8, 9, 11, 12, 13, 14, 16.

Interval (Rebounds)	Frequency
0–4	
5–9	
10–14	
15–19	

SHOW YOUR WORK

PART 2 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

A histogram of test scores shows: 50–59: 2, 60–69: 5, 70–79: 10, 80–89: 8, 90–99: 3. Which interval has the most students?

(A) 60–69 with 5 students

(B) 70–79 with 10 students

(C) 80–89 with 8 students

(D) 90–99 with 3 students

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A histogram of goals-per-game shows: 0–1: 9 players, 2–3: 6, 4–5: 3, 6–7: 1. What shape is this distribution?

(A) Perfectly symmetric

(B) Skewed right (most players low, tail toward high values)

(C) Flat / uniform (all bars equal)

(D) Skewed left (most players high, tail toward low values)

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The tallest bar is the 10–19 interval with 16 players. What does the shape of this histogram tell you about scoring in the league?

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LESSON 2.2

Represent and Describe Data in a Histogram

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Match each scenario to its histogram shape.

Groups: **Symmetric (peak in the middle)** **Skewed Right (tail points to high values)** **Skewed Left (tail points to low values)**

_____ Player heights on a team — most are average height, fewer are very tall or very short

_____ Points per game — most players score low, a few stars score very high

_____ Free-throw % of experienced players — most are high, a few are very low

2 **Circle the letter of the best answer. Show how you know.**

Two histograms show the SAME 40 test scores — one uses intervals of 5, the other intervals of 20. Which statement is true?

(A) The interval-of-5 histogram has more bars and shows more detail about the shape

(B) The interval-of-5 histogram leaves some scores out

(C) The interval-of-20 histogram has more bars

(D) Both histograms must look exactly the same

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A histogram of points-per-game has its tallest bars on the LEFT (low scores) and a long tail stretching RIGHT toward a few high scorers. How is this distribution described?

(A) Skewed left

(B) Symmetric

(C) Bimodal (two equal peaks)

(D) Skewed right

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Find the mistake. Explain it, then show the correct work.**

Find the Histogram Error

1. **Data** Test scores: 55, 62, 68, 70, 72, 75, 78, 80, 82, 85, 88, 90, 95
2. **Intervals** 50–59, 60–69, 70–79, 80–89, 90–100
3. **Frequency table** 50–59: 1, 60–69: 2, 70–79: 4, 80–89: 4, 90–100: 2
4. **Conclusion** The histogram is symmetric because the bars go 1, 2, 4, 4, 2.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

5 **Answer in complete sentences.**

A teacher collected test scores and made two different histograms — one with intervals of 5 (50–54, 55–59, etc.) and one with intervals of 20 (50–69, 70–89, 90–109). Both use the same data. How might the two histograms look different? Which interval size gives you more detail about the distribution? When might the larger interval be better?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

The tallest bar is the 10–19 interval with 16 players. What does the shape of this histogram tell you about scoring in the league?

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